# NETWORK SECURITY LABORATORY

## Experiment 2 – Configuring Switch Port Security

*Department: Cybersecurity Engineering Technologies | Fourth Year*

## 1. Main Objectives

- Understand the purpose of switch port security.
- Configure an access port and limit it to one MAC address.
- Use sticky MAC learning for the authorized computer.
- Detect and block an unauthorized computer.
- Recover a port disabled by a security violation.

## 2. Port Security in One Minute

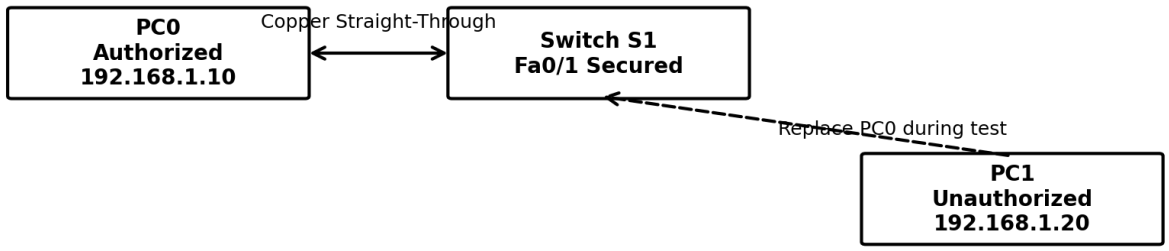
A switch normally learns MAC addresses automatically. Port Security restricts which MAC addresses are allowed to use an access port. In this lab, PC0 is authorized; when PC1 replaces it, the switch detects a different MAC address and shuts the port down.

**Key idea:** Sticky learning automatically learns the first authorized MAC address and stores it as a secure MAC address.

| Term | Meaning |
|---|---|
| Access Port | Connects an end device such as a PC. |
| Secure MAC | MAC address permitted on the protected port. |
| Sticky MAC | Automatically learned secure MAC address. |
| Violation | An unauthorized MAC attempts to use the secured port. |

## 3. Violation Modes

| Mode | Action |
|---|---|
| Protect | Drops unauthorized traffic. |
| Restrict | Drops traffic and records the violation. |
| Shutdown | Disables the port after a violation (used in this lab). |




PC0 and PC1 use the same switch port at different times.

| Device | IP Address | Subnet Mask | Role |
|---|---|---|---|
| PC0 | 192.168.1.10 | 255.255.255.0 | Authorized |
| PC1 | 192.168.1.20 | 255.255.255.0 | Unauthorized |

## 4. Build and Configure the Network

Add one Cisco 2960 switch and two PCs in Packet Tracer.

Connect PC0 FastEthernet0 to Switch0 FastEthernet0/1 using Copper Straight-Through.

Do not connect PC1 yet.

Configure PC0 as 192.168.1.10/24 and PC1 as 192.168.1.20/24. Leave Default Gateway empty.

## 5. Initial Switch Setup

```
enable
configure terminal
hostname S1
```

## 6. Configure Port Security on Fa0/1

```
interface fastEthernet 0/1
switchport mode access
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
no shutdown
end
```

| Command | Purpose |
|---|---|
| switchport mode access | Makes Fa0/1 an access port. |
| port-security | Enables port security. |
| maximum 1 | Allows one secure MAC address. |
| mac-address sticky | Learns the connected device MAC automatically. |
| violation shutdown | Disables the port when another MAC is detected. |

## 7. Learn the Authorized MAC

Generate traffic from PC0 so the switch can learn its MAC address:

```
ping 192.168.1.20
```

The ping may fail because PC1 is not connected, but PC0 still sends traffic and the switch can learn its MAC address.




Authorized MAC → Secure-up | Unauthorized MAC → Secure-shutdown / err-disabled

## 8. Verify the Authorized Device

```
show port-security interface fastEthernet 0/1
show port-security address
```

| Expected Item | Expected Value |
|---|---|
| Port Security | Enabled |
| Port Status | Secure-up |
| Violation Mode | Shutdown |
| Maximum MAC Addresses | 1 |
| Sticky MAC Addresses | 1 |
| Violation Count | 0 |

## 9. Test an Unauthorized Device

Disconnect PC0 from Fa0/1.

Connect PC1 to the same Fa0/1 port.

From PC1, generate traffic using: ping 192.168.1.10

The switch sees a different MAC address. With shutdown mode, Fa0/1 is disabled.

## 10. Check the Violation

```
show port-security interface fastEthernet 0/1
show interfaces status
```

| Item | Expected Result |
|---|---|
| Port Status | Secure-shutdown |
| Violation Count | 1 |
| Interface State | May appear err-disabled |

**Key idea:** err-disabled means the switch automatically disabled the interface because of an error or security violation.

## 11. Recover the Port

Disconnect PC1, reconnect PC0, then enter:

```
enable
configure terminal
interface fastEthernet 0/1
shutdown
no shutdown
end
show port-security interface fastEthernet 0/1
```

Expected state after recovery: Secure-up.

## 12. Save the Configuration

```
copy running-config startup-config
```

## 13. Quick Troubleshooting

| Problem | Check / Fix |
|---|---|
| Port security command fails | Configure switchport mode access first. |
| Sticky MAC does not appear | Generate traffic from PC0, then run show port-security address. |
| Port remains red | Reconnect PC0 and apply shutdown then no shutdown. |
| PC1 is not blocked | Verify Port Security: Enabled and Maximum MAC Addresses: 1. |

## 14. Student Practical Task

Create one switch and two PCs.

Configure PC0 as the authorized device on Fa0/1.

Enable port security, maximum 1, sticky learning and shutdown mode.

Generate traffic and verify the sticky MAC.

Replace PC0 with PC1 and generate traffic.

Observe the violation, reconnect PC0 and recover the port.

Save the Packet Tracer file as Week2_PortSecurity_StudentName.pkt.

Required screenshots: topology, port-security configuration, learned sticky MAC, secure-up before the violation, secure-shutdown after PC1, and violation counter.

## 15. Review Questions

**1. What is the purpose of Port Security?** Control which devices may use a switch port based on MAC addresses.

**2. What does maximum 1 mean?** Only one secure MAC address may use the port.

**3. What is sticky learning?** The switch automatically learns the connected MAC as a secure address.

**4. What happens in shutdown mode?** The switch disables the port when an unauthorized MAC is detected.

**5. How is a disabled port recovered?** Remove the unauthorized device, reconnect the authorized one, then use shutdown and no shutdown.

**6. Which command displays secure MAC addresses?** show port-security address

## 16. Expected Result Summary

| Test | Expected Result |
|---|---|
| PC0 connected | Secure-up |
| PC0 sends traffic | MAC learned as SecureSticky |
| PC1 replaces PC0 | Security violation |
| PC1 sends traffic | Port disabled |
| Violation counter | 0 → 1 |
| PC0 reconnected | Port can be manually recovered |